

CHAPTER 14 SUMMARY: DESCRIPTION OF MOTION

Essential Review for Grade 10 Physics

1. THE FUNDAMENTALS

- **Relative Motion:** Rest and motion depend on the **Frame of Reference** (Space + Time).
- **Trajectory:** The path of a mobile. Rectilinear = straight line; Curvilinear = curved line.
- **Position (x):** Determined by the abscissa $x = \overline{OM}$. (Unit: m).
- **Displacement (Δx):** Change in position $\Delta x = x_f - x_i$. (Can be positive or negative).
- **Duration (Δt):** Time interval $\Delta t = t_f - t_i$. (Unit: s).

2. VELOCITY AND SPEED

Average Speed (v_{av}): Ratio of total distance to total time. $v_{av} = \frac{d}{\Delta t}$

- **Instantaneous Speed (v):** Speed at a specific moment. Measured by speedometer or chronophotography.
- **Conversion:** $1 \text{ m/s} = 3.6 \text{ km/h}$ | $1 \text{ km/h} = \frac{1}{3.6} \text{ m/s}$.

3. VECTOR REPRESENTATIONS

Characteristic	Velocity Vector ($\vec{v}$)	Acceleration Vector ($\vec{a}$)
Origin	Current position M	Current position M
Line of Action	Rectilinear path (trajectory)	Rectilinear path (trajectory)
Direction	Direction of motion	Same as $\vec{v}$ (Acc.) or opposite (Decel.)
Magnitude	Instantaneous speed v	Instantaneous acceleration a

4. ACCELERATION & NATURE OF MOTION

- **Average Acceleration (a_{av}):** $a_{av} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$ (Unit: m/s^2).
- **Instantaneous Acceleration (a):** Rate of change of velocity at instant t .

Motion Classification

- **Uniform Rectilinear (URM):** v is constant $\Rightarrow a = 0$.
- **Accelerated Rectilinear:** $\vec{v}$ and $\vec{a}$ same direction ($v \cdot a > 0$). Speed is *increasing*.
- **Decelerated Rectilinear:** $\vec{v}$ and $\vec{a}$ opposite direction ($v \cdot a < 0$). Speed is *decreasing*.

5. CHRONOPHOTOGRAPHY FORMULAS (τ = INTERVAL)

For a recording of points $M_0, M_1, M_2 \dots$ separated by constant time τ :

a) **Speed at M_i :** $v_i = \frac{M_{i-1}M_{i+1}}{2\tau}$

b) **Acceleration at M_i :** $a_i = \frac{v_{i+1} - v_{i-1}}{2\tau}$

Note: Always ensure units are converted to SI ($m, s, m/s$) before calculation.